

AVRA PAUL

Software Developer | Web Developer | Machine Learning & Data Science

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

SUMMARY

Data Science undergraduate at IIT Madras who designs and ships end-to-end software, web, and machine learning applications. Hands-on experience building full-stack web apps (React.js, Node.js, Flask), RESTful APIs, ETL data pipelines, and LLM-powered AI agents, deployed to production with Git, Docker, and CI/CD. Seeking Software Engineer, Web Developer, or Machine Learning internship roles to build scalable, data-driven products.

EDUCATION

B.S. in Data Science & Applications, Indian Institute of Technology (IIT) Madras	CGPA: 8.43	2027
B.Tech in Information Technology, University of Kalyani	CGPA: 8.2	2027
Higher Secondary (HSC), Kalyani Public School	82%	2023
Secondary (SSC), St. Claret School	91%	2021

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS, TypeScript
Frontend: React.js, HTML5, CSS3, Responsive Web Design, RESTful API Integration
Backend: Node.js, Flask, RESTful API Development, Express.js
Databases: MySQL, PostgreSQL, SQLite
Machine Learning & Data Science: Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, Exploratory Data Analysis (EDA), Feature Engineering, Model Training & Evaluation, NLP
AI / LLM: LLM Application Development, Retrieval-Augmented Generation (RAG), OpenAI API, LangChain, Prompt Engineering
Data Visualization: Matplotlib, Seaborn, Plotly
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design Basics, API Design
Tools & Platforms: Git, GitHub, Docker, Postman, Playwright, Jupyter Notebook, Google Colab
Cloud & DevOps: Vercel, Render, CI/CD, API Deployment, Version Control

PROJECTS

- Data Analyst Agent — Alvyn v2** <https://alvyn-v2.vercel.app> | *Python, Flask, OpenAI API, Plotly*
- Built an end-to-end AI data analyst agent that accepts natural language queries, autonomously scrapes and cleans data, and returns structured JSON insights with base64-encoded visualizations
 - Designed ETL pipelines for dynamic data ingestion; exposed a RESTful API consumed by the React frontend
 - Deployed on Vercel with scalable serverless architecture; integrated OpenAI API for LLM-driven query resolution
- Distributed Job Queue & Observability Dashboard** | *Python, FastAPI, Redis, PostgreSQL, WebSockets, Docker*
- Designed a distributed task queue using Redis and async Python workers, supporting concurrent job execution with priority scheduling and dead-letter retry logic
 - Streamed real-time job progress to a React dashboard via WebSockets, replacing client-side polling with server-push updates
 - Containerized all services (API, workers, Redis, Postgres) with Docker Compose for reproducible single-command deployment
- Multi-Class Comment Classification** [Github](#) | *Python, scikit-learn, Pandas, TFIDF, NLP*
- Built and evaluated multi-class text classification models (4 classes: hate, neutral, supportive, disrespect) on Kaggle competition dataset
 - Achieved 85%+ accuracy through feature engineering: TFIDF vectorization, n-gram analysis, and text preprocessing (tokenization, lowercasing, stopword removal, word overlapping)

- Experimented with multiple classifiers (Logistic Regression, SVM, Random Forest, XGBoost, CatBoost) and stacking models; selected best performer via cross-validation and grid search

Business Performance Analytics — Retail Enterprise | *Python, Pandas, Matplotlib, Scikit-learn*

- Analyzed multi-dimensional operational, cost, and revenue datasets to identify profitability and efficiency trends
- Built pricing and cost-structure models; surfaced insights that informed inventory and pricing strategy decisions

EXPERIENCE

Data Science & AI Intern, ConglomerateIT (On-Site)

3 months

- Automated hiring workflows using Python, reducing manual processing time by ~60%.
- Performed data analysis to identify bottlenecks and improve recruitment efficiency.
- Collaborated with cross-functional teams to design and implement data-driven solutions.

Data Science & AI Intern, CoLab (Remote)

2 months

- Owned data collection, cleaning, and analysis to support data-driven decision-making.
- Updated the company website and strengthened the backend for improved robustness and reliability.

Co-Founder & Full-Stack Developer, BrewWeb Agency

brew-web.vercel.app

- Delivered full-stack web applications for local cafes and restaurants, covering frontend integration and backend API design.
- Collaborated directly with clients to translate business requirements into working product features.
- Built and deployed production-ready applications using RESTful services.

STRENGTHS

- Strong data engineering and software fundamentals with real-world scraping and pipeline experience.
- Clear communicator able to translate complex data into actionable insights.
- Thrives in fast-paced, problem-driven environments and ships end-to-end.

ADDITIONAL INFORMATION

Languages: Bengali, English, Hindi